Binomial Series

Questions 1–23

Question 1. For each of the following, expand up to and including the x^3 term, stating in each case the values of x for which the expansions are valid:

a) $(1+x)^{\frac{1}{4}}$
b) $(1+2x)^{-1}$
c) $\sqrt{1-3x}$
d) $(2+x)^{-3}$
e) $(9+3x)^{\frac{1}{2}}$
f) $\sqrt{16-36x}$
g) $\sqrt[3]{8-2x}$
h) $\frac{1}{\sqrt{3+2x}}$

Question 2. Write down the coefficient of the x^4 term in each of the following expansions:

a) $(1+3x)^{\frac{1}{4}}$
b) $(1-2x)^{\frac{2}{3}}$
c) $(1+\frac{x}{2})^{-\frac{3}{4}}$

Question 3.

a) Find the first 3 terms in the expansion of $\frac{1}{\sqrt{9+x}}$

b) Hence find the first 3 terms in the expansion of $\frac{1}{\sqrt{9-x^2}}$

Question 4.

a) Find the expansion of $(1+x)^{-\frac{1}{2}}$ up to and including the x^2 term.

b) Hence find the first 3 terms in the expansion of

$$(1+\tfrac{2}{5}x)^{-\frac{1}{2}} \quad \text{and} \quad \sqrt{\frac{10}{5+2x}}$$

Question 5.

a) Find the first 3 terms in the expansion of $\sqrt{9-x}$

b) Hence find an approximation to $\sqrt{8}$

Question 6.

a) Expand $(1+5x)^{\frac{1}{3}}$ up to and including the term in x^3.

b) Hence find an approximation to the value of $\sqrt[3]{1.1125}$ to 4 decimal places.

Question 7. By finding the first 3 terms in the expansion of $(2-3x)^{\frac{1}{2}}$ use $x=\frac{1}{10}$ to find an approximation for $\sqrt{170}$ giving your answer in the form $p\sqrt{2}$ where p is a value to be found.

Question 8.

a) Find the first 3 terms in the expansions of $(1+3x)^{\frac{1}{2}}$ and $(1+x)^{-\frac{1}{2}}$

b) Hence, show that

$$\sqrt{\frac{1+3x}{1+x}} \approx 1+x-\frac{3}{2}x^2$$

c) State the values of x for which the expansion is valid.

Question 9.

a) Find the first 3 terms in the expansion of $(1-2x)^{\frac{1}{2}}$

b) Hence, show that

$$\frac{\sqrt{1-2x}}{1+x} \approx 1 - 2x + \frac{3}{2}x^2$$

c) State the values of x for which the expansion is valid.

Question 10. Given the binomial expansion

$$(1+qx)^p = 1 - 2x + 6x^2 + \dots$$

where p and q are constants.

a) Find the values of p and q.

b) Hence state the values of x for which the expansion is valid.

Question 11. Given that

$$\sqrt[3]{8-x} \approx 2 - \frac{1}{12}x + kx^2 + \dots$$

where k is a constant.

a) Find the value of k.

b) By using $x = 1$ in the expansion, find an approximation to the value of $\sqrt[3]{7}$.

Question 12.

a) By using $x = \frac{1}{8}$ in the first three terms in the expansion of $\sqrt{3-8x}$, find an approximation for the value of $\sqrt{2}$.

b) Explain whether your approximation is an under- or over-estimate.

Question 13.

a) Find the terms up to and including the x^2 term in the expansion of $\frac{1}{\sqrt{2-x}}$

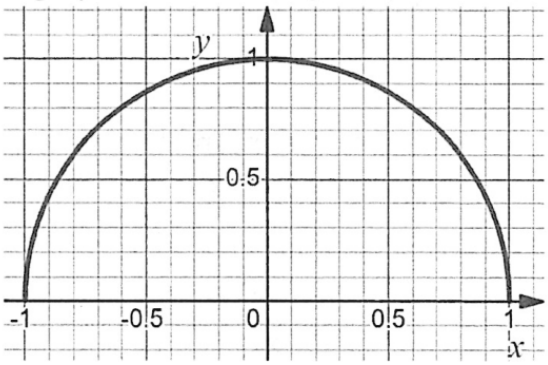
Jenna considers using the values $x = -3$, $x = -\frac{6}{5}$ and $x = \frac{1}{5}$.

b) Explain how she used the values to find $\sqrt{5}$.

c) Which of the three values should not be used in the expansion?

Question 14. By considering the first three terms in the expansion of $\frac{1}{\sqrt{7-x}}$, find an approximation for the value of $\sqrt{3}$.

Question 15. The graph below shows $y = \sqrt{1 - x^2}$.



By considering the graph of $1 + \frac{1}{2}x^2 + \ldots$ explain why this is not a correct expansion of $\sqrt{1 - x^2}$.

Question 16. Find the values of a and b given

$$\frac{a + bx}{\sqrt{9 - x}} = 1 - \frac{11}{18}x - \frac{7}{216}x^2$$

where a and b are constants.

Question 17.

a) Find the first 4 terms in the expansion of $(1+px)^{\frac{1}{2}}$ where p is a constant

Given for the constant term q

$$(1+qx)(1+px)^{\frac{1}{2}} = 1 - 4x - 14x^2 + 16x^3 - \dots$$

b) find the values of p and q

Question 18.

a) Find the first 4 terms in the expansion of $(1-x)^{-2}$

b) Hence find $2 + 3x + 4x^2 + 5x^3 + \dots$ in the form $\frac{a-x}{(b-x)^2}$ where a and b are integers to be found.

Question 19. Prerequisite knowledge for Q19–Q21: Partial Fractions It is known that

$$\frac{2-5x}{(x-1)(2x-1)}, \qquad x \neq 1,\ x \neq \frac{1}{2}$$

can be expressed in the form $\frac{A}{x-1} + \frac{B}{2x-1}$.

a) Find the values of A and B.

b) Hence find the first three terms in the expansion of $\frac{2-5x}{(x-1)(2x-1)}$.

Question 20. By expressing

$$\frac{11x+5}{(5x+1)(x+3)}, \qquad x \neq -\frac{1}{5},\ x \neq -\frac{1}{3}$$

in the form $\frac{A}{5x+1} + \frac{B}{x+3}$

a) Find the first 3 terms in the binomial expansion of $\frac{11x+5}{(5x+1)(x+3)}$.

b) State the values of x for which the expansion is valid.

Question 21. By expressing

$$\frac{9x^2 - 5x + 2}{(x+1)(3x+1)^2}, \qquad x \neq -1,\ x \neq -\frac{1}{3}$$

as partial fractions,

a) Find the first 3 terms in the binomial expansion of $\frac{9x^2-5x+2}{(x+1)(3x+1)^2}$.

b) State the values of x for which the expansion is valid.

Question 22. Prerequisite knowledge for Q22: Integration of Polynomials

a) Find the first three terms in the expansion of $\frac{1}{\sqrt{9-x}}$

b) Hence find an approximation for $\int_0^1 \frac{1}{\sqrt{9-x^2}}\,dx$

Question 23. Prerequisite knowledge for Q23: Radians A 1 radian sector of a circle has area $(x+2)$ units^2. Show that for small values of x, the radius is given by

$$r \approx 2 + px - qx^2$$

where p and q are constants to be found.